Unit 2 - Handout 3
The Pumping Lemma for Regular Languages

If L is regular, there is a constant p (the pumping length)
such that every string w in L with |w| >= p can be split
as w = xyz where:
  1. |xy| <= p
  2. |y| >= 1
  3. x y^i z is in L for every i >= 0

Using it to prove L = { a^n b^n : n >= 0 } is not regular:
Assume L is regular with pumping length p. Take w = a^p b^p.
Since |xy| <= p, y consists only of a's. Pumping with i = 2
gives more a's than b's, which is not in L. Contradiction.

Note: the lemma is a necessary condition only. It can show
that a language is NOT regular, never that it IS regular.

Formal Languages and Automata